

Taylor

$$f(a) + \frac{f'(a)}{1!}(x-a) + \dots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Nabla Operator

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

$$\nabla f(x) = A x + b$$

$$\nabla^2 f(x) = \Lambda$$

$$x = [x_1, x_2, x_3]^T$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$x \in \mathbb{R}$$

$$\text{Symmetry: } \frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

$$A^T y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 22 \end{bmatrix}.$$

$$(A^T A)^H = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$$

$$Q \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 10 \\ 22 \end{bmatrix} = \begin{bmatrix} 0.617 \\ 1.8 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 13.1 & 619 & 23 \\ 1 & 13.5 & 803 & 42 \\ 1 & 13.8 & 207 & 31 \\ 1 & 13.1 & 409 & 38 \\ 1 & 12.5 & 482 & 45 \end{bmatrix}$$

$$y = [181 \quad 192 \quad 212 \quad 221 \quad 248].$$

$$A^T \cdot A =$$